New sweep, z = 0 mm: motor-to-sensor identification

Independently unwrapped; common 50 Hz branch convention; no smoothing. Each curve uses its own interpolated 50 Hz phase in [-270,90) deg.

Closed-loop total-input/output fit; first/second segments of the same sweep.

Check-segment FRF SRIVC model

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Unwrapped phase (deg)

Frequency (Hz)